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BEHAVIOR OF SINGLE PILE IN COHESIONLESS SOIL ON HORIZONTAL & SLOPING GROUND SURFACE UNDER LATERAL LOADING

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Abstract

Piles, ubiquitous in construction and structural engineering, serve a crucial role in supporting various structures. It represents a common foundation solution that offers essential support for a wide range of structures. Understanding the behavior of piles is paramount for ensuring the stability and integrity of these structures. Despite their widespread use, there remains a need for a deeper understanding of pile behavior, particularly in sloping ground condition. This research presents a small-scale experimental investigation on a single pile installed on both horizontal and sloping ground surface. The experiment analyzed piles with different length to diameter (L/D) ratio under two types of loading conditions such as forward and reverse lateral loading on sloping ground surface. The study further explores how pile behavior varies between mild and steep slopes in comparison to horizontal ground. The load carrying capacity of pile appeared to be improved with an increase of L/D ratio while steeper slope seems to reduce the load carrying capacity of pile compared to mild slope. The load-carrying capacity is further influenced by the loading conditions, with forward loading resulting in a certain degree of deflection at a smaller load than reverse loading, which acts against the slope. The discoveries made in this study are anticipated to make a substantial impact in the field, providing a general perspective on the interaction between soil and structure under diverse circumstances. Moreover, gaining insight into these interactions at a core level could lay the groundwork for implementation in construction and structural engineering, potentially driving the innovation of more resilient and sustainable structural systems.

Keywords: Pile foundation; Sloping condition; Lateral load; Soil-structure interaction; Forward and reverse loading

1 Introduction

A pile is a structural component frequently utilized in civil engineering and construction particularly for deep foundations or foundations in challenging soil conditions or to support heavy loads for a variety of structures. Essentially, it is a long, slender column, typically composed of concrete, steel, or timber.

A pile, employed in the construction of a structure, may be singular or a group of piles as well as can be classified as short or long based on the length-to-diameter ratio. Single pile is utilized in scenarios necessitating localized support, while group piles distribute loads over a wider area, making them ideal for larger structures or sites with variable soil conditions. On the other hand, short piles typically experience less lateral displacement, whereas long piles, by contrast, exhibit significant lateral displacement. This distinction is crucial in determining the appropriate pile type for a given structural requirement, ensuring stability and longevity of the foundation.When considering pile behavior, type of pile to be installed, the external forces such as wind loads, earth pressures, and seismic activity that acts laterally on them and the ground condition is very crucial as the behavior largely varies due to these factors. Unlike shallow foundations, deep piles embedded further into the ground can effectively withstand the abovementioned lateral forces. Moreover, in cases where a weak soil layer is present beneath the structure, piles offer support to structures with depths significantly exceeding their widths.

Structures, where single pile is employed, confronts immense lateral loads from wind and seismic activity and as a result, it undergoes deflections and experiences bending moments throughout its length. The lateral resistance of a single pile hinges upon several factors, including its diameter, the characteristics of the surrounding soil, the gradient of the ground surface, and the nature of the applied loads. In granular soils, the primary source of lateral resistance arises from soil arching and passive resistance. Any soil displacement compromising the interaction between soil and structure poses a significant risk to the stability of the superstructure. Therefore, it is imperative to understand and mitigate these potential risks.

Several studies have investigated the impact of slope angles and soil conditions on the lateral load capacity of piles. Research shows that piles on sloping ground have reduced lateral capacity, particularly near the slope crest, with the effect diminishing as the distance from the crest increases. Submerged soil conditions also lead to greater pile deflections and reduced load capacity, with soil density playing a critical role in pile behavior [1–3]. Additionally, the importance of pile-soil interaction under lateral loads has been explored using advanced methods like the Fourier Finite Element Method (FEM) and 3D analysis to develop accurate models for predicting pile deflection and bending moments, emphasizing the need to consider soil profiles to optimize pile design [4–6]. Further research on pile groups in marine and cohesionless soils has shown that factors such as pile rigidity, embedment length, and arrangement significantly influence lateral load capacity, with critical spacing between piles and alternative approaches to traditional methods like the p-y method offering practical solutions for pile design [7–10].

This study investigates the lateral load-carrying capacity of piles under both forward and reverse loading conditions on horizontal and sloping ground surfaces. By analyzing the influence of the length-to-diameter (L/D) ratio, it aims to understand the overall response of piles. Comparing load-carrying capacitiy between different sloping ground surfaces provides valuable insights into the effects of inclination on pile behavior in cohesionless soil.

Pile foundations on sloping surfaces present significant challenges in geotechnical engineering, particularly in areas with varied topographies like hilly regions and coastal zones. The behavior of piles on sloping ground differs from that on flat surfaces due to distinct load distribution and soil interactions. Despite the importance, research on sloping conditions is limited comared to the horizontal ground surface. This study examines the relationships between load, deflection, fixity depth, embedment length, and L/D ratio, offering crucial insights into pile performance in such ground surfaces.

2. Materials and Methods

2.1 Material Properties

2.1.1 Soil Properties

Table 1: Soil Properties

Properties	Value
D_{10}	0.31 mm
D_{30}	0.42 mm
D_{60}	0.65 mm
Co-efficient of uniformity, C_u	2.097
Co-efficient of contraction, C_c	0.875
Specific gravity, G_s	2.66
Relative density	95%
Internal friction angle, φ	35°

2.1.2 Pile Properties

Table 2: Properties of stainless-steel model pile

Properties	Pile 1	Pile 2
Diameter, D (mm)	25	25
Embedded length, L (mm)	580	650
Length to diameter ratio, L/D	23	26
Modulus of elasticity, E (kN/m^2)	195×10^6	195×10^6
Moment of inertia, I (m^4)	4.46×10^{-9}	4.46×10^{-9}

2.2 Methodology

2.2.1 Experimental Setup

To determine the required tank size, the pile's influence zone using Rankine's earth pressure theory was calculated first. A η_h value of 7500 kN/m³ for medium sand was taken into consideration and then stiffness factor of the pile is determined by using the values from Table 2. Additionally, the depth of fixity 'Z_f' is utilized to calculate the zone of influence of the pile and the value of internal friction angle from Table 1.

Stiffness Factor, $T = (\frac{EI}{\eta_h})^{\frac{1}{5}} = 0.163$

Depth of fixity (for horizontal ground), $Z_f = 1.85T = 0.302$

Influence of passive zone= $Z_f * \tan\left(45^0 + \frac{\phi}{2}\right) = 580mm$

Hence, the tank size of 0.73 m × 0.73 m × 0.73 m was selected to carry out the experimental work. A rectangular concrete box was used for this experiment as shown in [Fig. 1]. A vertical stand made of cast iron and a pulley system was constructed to transfer horizontal loads to the model pile. To prevent the stand from overturning during loading, a reinforced concrete pedestal was cast and the stand was embedded within it.

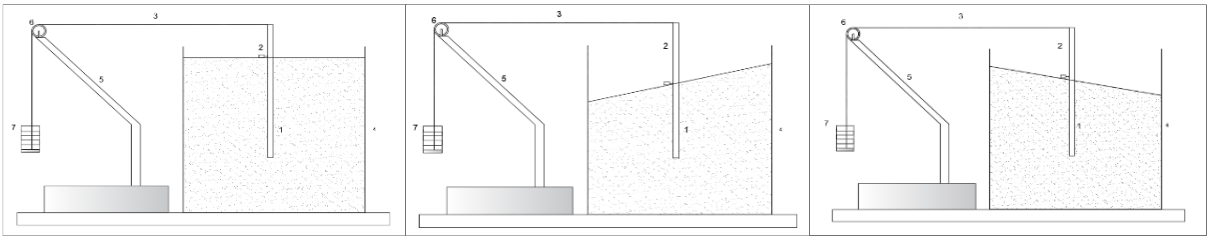

Figure 1: Experimental setup with various ground condition and loading arrangements

2.2.2 Sample Preparation

In this small-scale model, cohesionless granular soil was used. Compacted dry unit weight of 95% of the maximum dry unit weight of the soil sample, which was determined in the laboratory by the modified proctor test, was obtained. From [Fig. 2]. 95% of the maximum dry unit weight is $\gamma_d = 17$ kN/m³. As the moisture content at 95% of the maximum dry unit was 9.9% & field moisture content was 0.38%, to obtain the 95% relative compaction in the sample, 9.52% moisture was added.

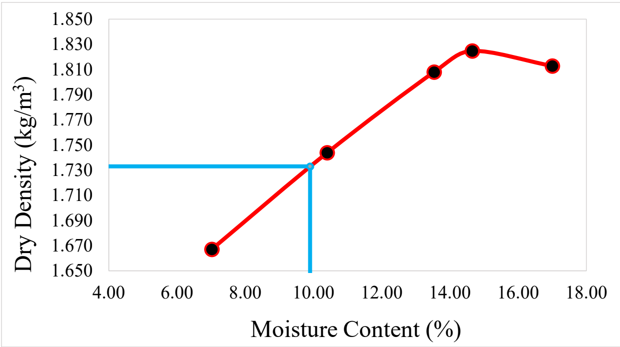

Figure 2: Dry density vs Moisture content by Modified proctor test

Void ratio, $e = \frac{G_s \gamma_w}{\gamma_d} - 1 = 0.535$

Volume of solid, $V_s = \frac{V_T}{1+e} = 0.23$ m³

Weight of solid, $W_s = V_s G_s \gamma_w = 6$ kN

Amount of water to be added was $(6 \times 9.52\%)kN = 58.23L$. The sand was filled in the concrete box upto a height of 0.68 m and a total of 5 layers, ensuring each layer received the same amount of soil and the same number of blows to maintain similar density in all the layers. As a result, each layer received 11.65 liters of water.

2.2.3 Slope formation

Initially, a horizontal ground surface of soil was established as a baseline with a height of 68 cm from the inner bottom of the tank, satisfying the zone of influence of the pile. Subsequently, a mild slope was created with the configuration of 1V:6H and later with the configuration of 1V:3H. Construction involved careful grading and shaping of the ground to achieve the desired slope angles.

2.2.4 Loading Arrangements

A pulley with a concrete block as base and counterweight was made. A 12-gauge GI wire was used for applying loads. A hook stand was hung on one end of the wire and the other end was tied to the pile. Incremental load was added to the hook stand, which in turn applied lateral load to the pile as shown in [Fig. 3]. This process was repeated until either the pile failed or the maximum deflection value of 50% of the pile diameter was reached. Load blocks weighing 4kg and 8kg were utilized to apply the lateral load.

A horizontal surface & two different slopes (1V:6H, 1V:3H), and two types of loading, such as pull along the slope and pull against the slope, were considered in this study along with piles of two different length to diameter ratio such as 23 and 26. The pull along and against the slope also can be called forward loading and reverse loading respectively.

Figure 3: Loading arrangements

3 Result & Discussion

A dial gauge of 0.01 mm accuracy was used for the collection of deflection values corresponding to the incremented lateral loadings. It was fixed just above the ground surface to measure displacement due to lateral load. Simultaneous loading and data collection were continued until the piles failed or exhibited considerable displacement.

3.1 Load-Displacement Response for Horizontal Ground Surface

Matlock & Reese (1960) provided a theoretical equation for displacement of pile at any given depth[11]. At a depth z from the ground surface, the displacement of pile can be mathematically expressed as-

$$X_{z(z)} = A_x \frac{Q_g T^3}{E_p I_p} + B_x \frac{M_g T^2}{E_p I_p} \qquad (1)$$

For horizontal ground condition, experimental result for both L/D ratio of 23 and 26 was compared to the theoretical results from Eq. (1). The theoretical and experimental results are quite similar as shown in [Fig. 4].

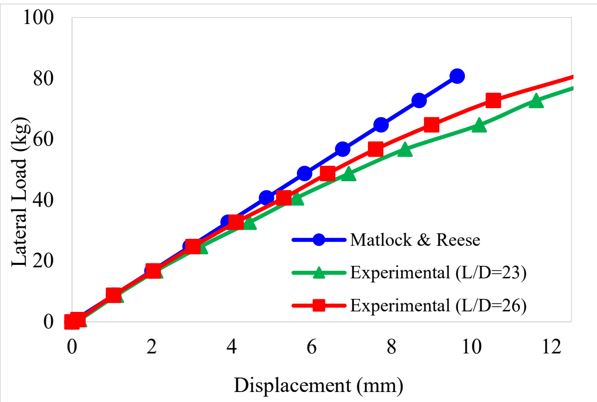

Figure 4: Load vs displacement curve in the horizontal ground for theoretical & experimental condition

3.2 Load-Displacement Response in Sloping Ground Surface

From [Fig. 5] and [Fig. 6], due to forward loading on a slope of 1V:6H, increasing the L/D ratio from 23 to 26, which is about 13%, raised the lateral load capacity by about 6% at the same deflection and reduces the deflection by 8% under ultimate loading. For reverse loading, on the same slope, a 5% increase in load capacity and an 8% deflection reduction with similar changes were found for a change of L/D ratio from 23 to 26.

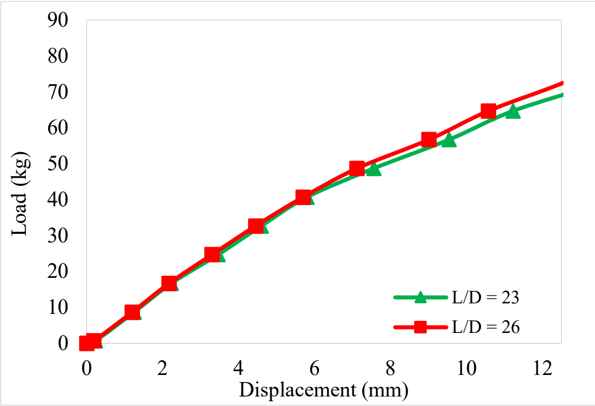

Figure 5: Load vs displacement curve in the sloping surface (1V:6H) for forward loading

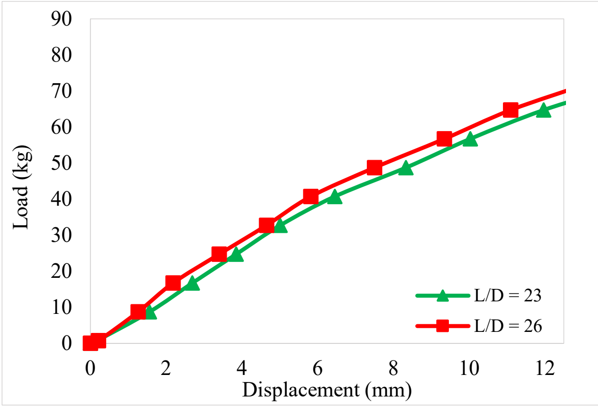

Figure 6: Load vs displacement curve in the sloping surface (1V:6H) for reverse loading

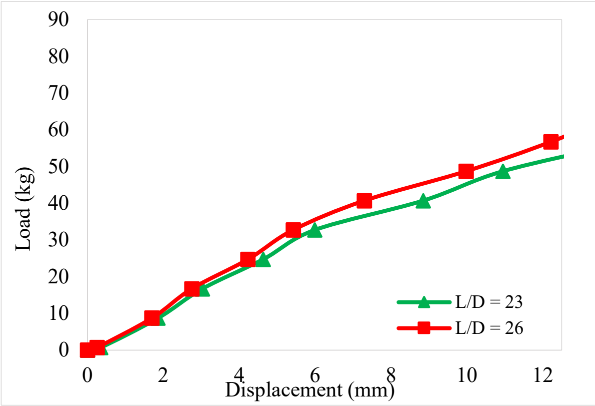

Figure 7: Load vs displacement curve in the sloping surface (1V:3H) for forward loading

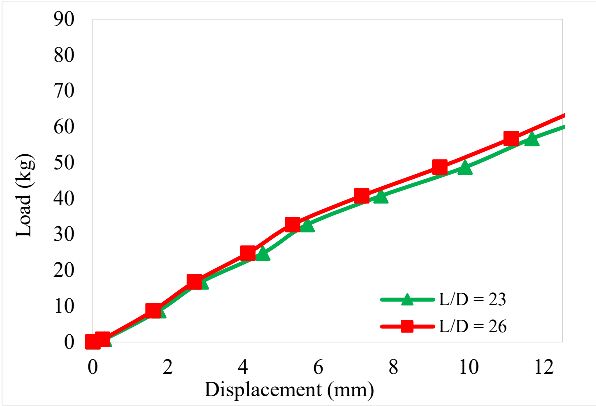

Figure 8: Load vs displacement curve in the sloping surface (1V:3H) for reverse loading

[Fig. 7] & [Fig. 8] demonstrate that, on a comparatively steeper slope of 1V:3H, increasing the L/D ratio from 23 to 26 resulted in a 10% increase of load carrying capacity at the same deflection and leads to a 17% reduction in deflection. Reverse loading causes an increase of about 5% in load carrying capacity and a 6% reduction in deflection while the other conditions stay the same.

3.3 Comparison of Lateral Load Capacity for 1V:6H Sloping Surface

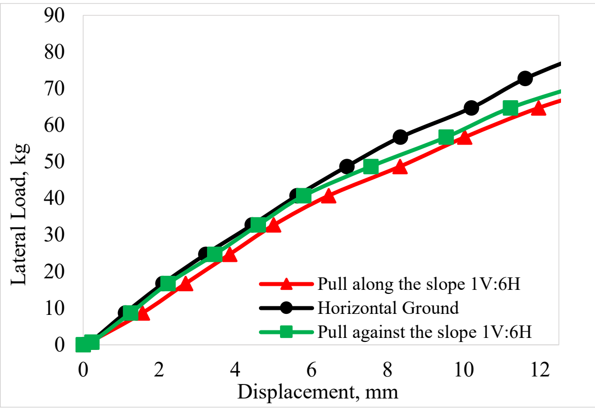

Figure 9: Comparison of lateral load capacity between horizontal & sloping (1V:6H) ground for L/D=23

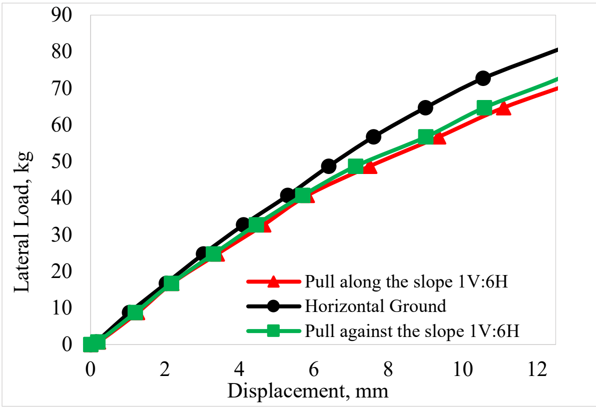

Figure 10: Comparison of lateral load capacity between horizontal & sloping (1V:6H) ground for L/D=26

[Fig. 9] & [Fig. 10] demonstrate a significant reduction in the load-carrying capacity of piles on sloping surfaces compared to horizontal ground. With an L/D ratio of 23 and a slope of 1V:6H, the capacity decreases by about 13% when pulled along the slope and 10% when pulled against it, compared to horizontal ground at ultimate deflection, which is 50% of pile diameter. Additionally, pulling along the slope results in a 5% lower capacity than pulling against it. On the other hand, for an L/D ratio of 26, similar trends were observed. The load-carrying capacity remained consistent under horizontal ground conditions and for pulls along and against the slope (1V:6H). But it was clearly experienced that, with increasing L/D ratio, the load carrying capacity was increased for every step from the previous one.

3.4 Comparison of Lateral Load Capacity for 1V:3H Sloping Surface

As shown in [Fig. 11] & [Fig. 12], with an L/D ratio of 23 and a slope of 1V:3H slope, pulling along the slope results in a 32% reduction in load-carrying capacity, while pulling against the slope leads to a 21% decrease, both compared to horizontal ground conditions at a deflection of 50% pile diameter. Additionally, pulling along the slope yields a 13.5% lower capacity than pulling against it. Similar trends were observed as the L/D ratio increased, consistent with earlier findings at an L/D ratio of 23, where load-carrying capacity steadily increased under horizontal and sloped conditions.

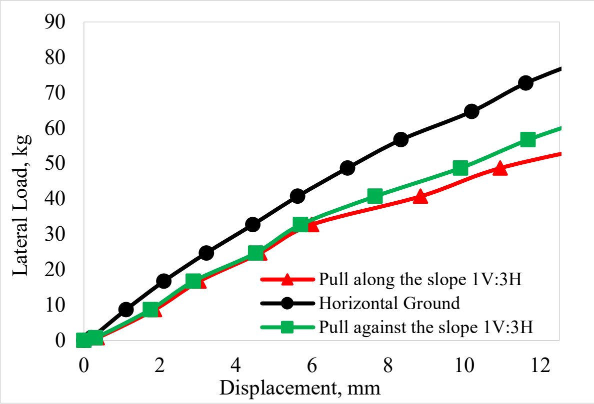

Figure 11: Comparison of lateral load capacity between horizontal & sloping (1V:3H) ground for L/D=23

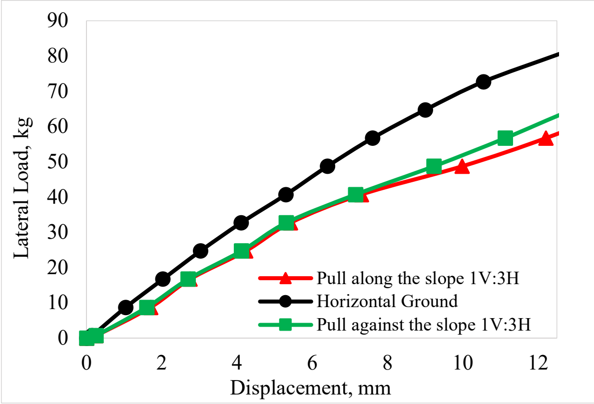

Figure 12: Comparison of lateral load capacity between horizontal & sloping (1V:3H) ground for L/D=26

3.5 Comparison of Lateral Load Capacity Under Forward Loading

The comparison of load-carrying capacities between horizontal and sloping ground (1V:3H, 1V:6H) under forward loading, with an L/D ratio of 23 as demonstrated in [Fig. 13], showed significant reductions of about 32% for 1V:3H and 13% for 1V:6H compared to horizontal conditions. This trend was consistent at an L/D ratio of 26 as demonstrated in [Fig. 14]. The findings highlight the critical impact of slope angle on pile behavior, with steeper slopes causing more significant reductions in load-carrying capacity.

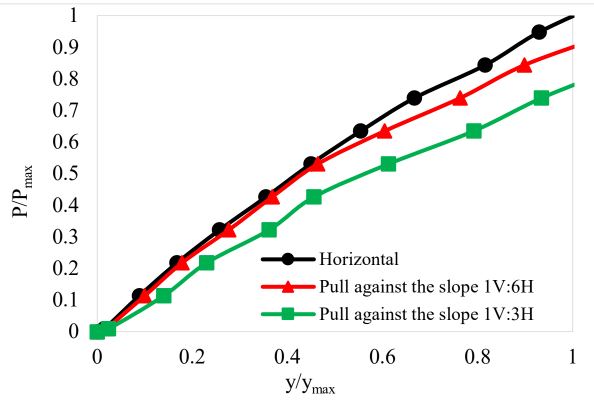

Figure 13: Comparison of lateral load capacity due to forward loading between horizontal & sloping ground for L/D=23

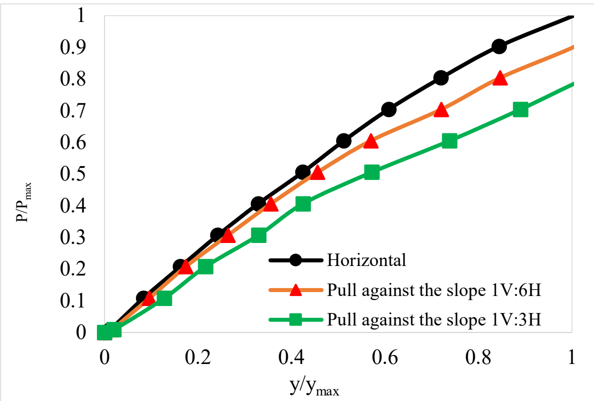

Figure 14: Comparison of lateral load capacity due to forward loading between horizontal & sloping ground for L/D=26

3.6 Comparison of Lateral Load Capacity Under Reverse Loading

The comparison between load capacities on flat and sloped ground (1V:3H, 1V:6H) revealed a decrease in load-bearing ability when subjected to forward loading as shown in [Fig. 15] & [Fig. 16]. With an L/D ratio of 23 or 26, the load capacity diminished by roughly 21% on the steeper slope (1V:3H) and 10% on the gentler slope (1V:6H) compared to flat ground. It was clearly observed that the load-bearing capacity exhibited a notable increase during reverse loading compared to forward loading.

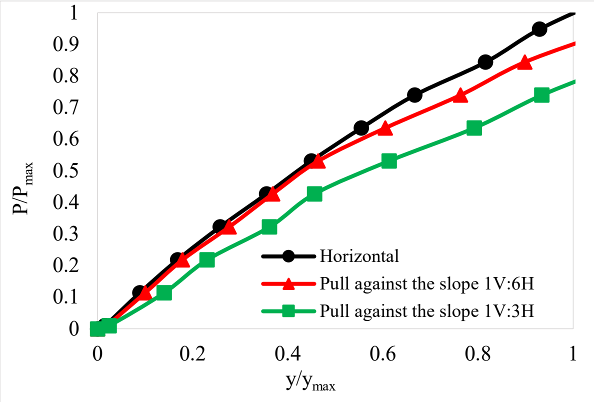

Figure 15: Comparison of lateral load capacity due to reverse loading between horizontal & sloping ground for L/D=23

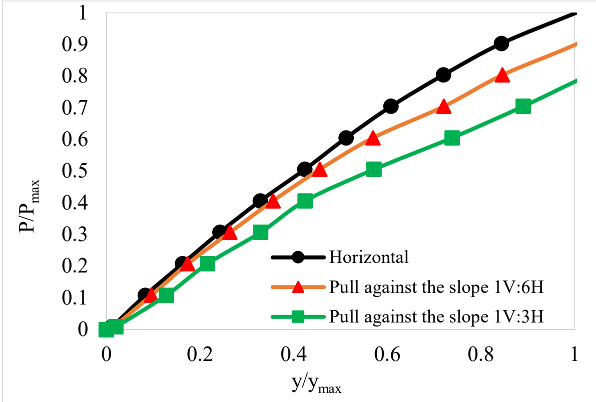

Figure 16: Comparison of lateral load capacity due to reverse loading between horizontal & sloping ground for L/D=26

Overall, these findings highlight the critical role of slope angle in determining the lateral response of piles and underscore the need for thorough consideration of slope conditions in structural design to ensure stability and performance.

4 Conclusion

The results of the experimental investigation in this study revealed that the change in slope on the ground surface has a significant impact on the pile's resistance to lateral loads. The lateral displacement of the pile is greatly reduced with changing of slope from mild to steep. In the context of the experimental results and analysis, the following conclusions are also drawn:

i. The direction of loading and change in the slope of ground surface significantly affect the lateral load capacity of the pile.

ii. For horizontal ground surface, as the L/D ratio increases, , the observed curve tends to approach Matlock & Reese's (1960) theoretical curve.

iii. Under forward loading, the lateral load capacity of the pile decreases by 13% and 32% for the slope of 1V:6H and 1V:3H, respectively, compared to horizontal ground conditions, having similar trends observed for L/D ratios of both 23 and 26.

iv. Conversely, under reverse loading, the lateral load capacity decreases by 10% and 21% for the slope of 1V:6 H and 1V:3H slopes, respectively, compared to horizontal ground conditions.

v. In forward loading, the lateral load capacity is 5% to 13.5% lower than in reverse loading, depending on the slope inclination (from 1V:6H to 1V:3H).

vi. Due to reduced lateral load capacity in sloping ground compared to horizontal ground, the depth of pile fixity is greater in sloping ground.

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